Kindly follow the **instructions** below:

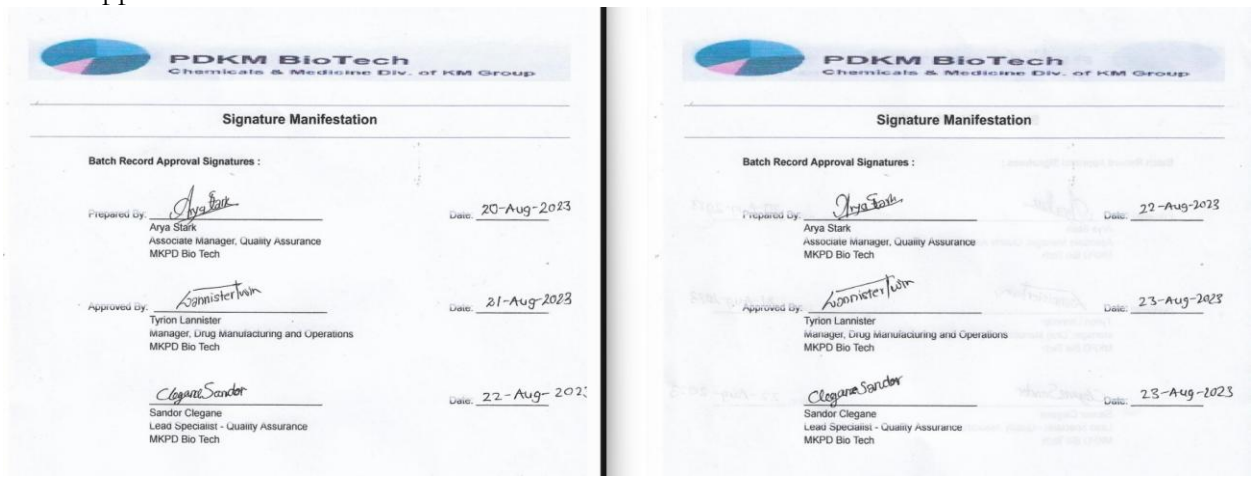
1. For assignment completion, you may choose any OCR tools, algorithms, or statistical methodologies you consider appropriate; there are no restrictions on the technical stack or approach.
2. Evaluation will primarily focus on the clarity, robustness, and depth of your chosen strategy, methodology, and overall approach as documented in a detailed report, rather than on final results, outcomes, or accuracy metrics alone.
3. Submit all source code, pipelines, and any modular Python packages in a zipped folder, along with a detailed report describing your approach, methodology, and results, on or before the due date and time highlighted in the email.
4. Solve one question out of Questions **1 and 2**.
5. Solve one question out of Questions **3 and 4**.
6. Questions **5 and 6** are **compulsory** and must be attempted.

Assignment Assessment

1. On Page No. 1 in both documents, Sample-9690.pdf and Sample-9700.pdf, 'Tyrion Lannister' and 'Sandor Clegane' approved the Batch Records report. Extract the 'Name', 'Designation', 'Company', and 'Date of Approval' for both approvers, save them in 2 separate tables (one for Sample-9690.pdf and another one for Sample-9700.pdf), and also save a third table in which the time difference (how long it took to approve the records for 'Tyrion Lannister' and 'Sandor Clegane') in approval of Sample-9700.pdf after Sample-9690.pdf was approved.

Note: You can save in either tabular/dictionary/JSON.

Ref: Snippet



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Chemicals & Medicine Div. of KM Group

Signature Manifestation

Batch Record Approval Signatures :

Prepared By: Arya Stark — Date: 20-Aug-2023
Arya Stark
Associate Manager, Quality Assurance
MKPD Bio Tech

Approved By: Lannister Tyrion — Date: 21-Aug-2023
Tyrion Lannister
Manager, Drug Manufacturing and Operations
MKPD Bio Tech

Clegane Sandor — Date: 22-Aug-2023
Sandor Clegane
Lead Specialist - Quality Assurance
MKPD Bio Tech

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Signature Manifestation

Batch Record Approval Signatures :

Prepared By: Arya Stark — Date: 22-Aug-2023
Arya Stark
Associate Manager, Quality Assurance
MKPD Bio Tech

Approved By: Lannister Tyrion — Date: 23-Aug-2023
Tyrion Lannister
Manager, Drug Manufacturing and Operations
MKPD Bio Tech

Clegane Sandor — Date: 23-Aug-2023
Sandor Clegane
Lead Specialist - Quality Assurance
MKPD Bio Tech

2. In Sample-9690.pdf, the 'Bill of Material' is present on Page No. 2, and in Sample-9700.pdf, the 'Bill of Material' is present on Page No. 3. Map these 2 pages and give the change in values for Magnesium Stearate, Sodium Lauryl Sulfate, Stearic Acid, and Titanium Dioxide.

Note: You can save results in a tabular/dictionary/JSON

Ref: Snippet

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

1. Bill Of Material

Serial Number	Material ID	Material Description	Lot Number	Quantity	Expiry Date	Issued By	Received By
1	A2R7F9	Acetaminophen	ST 8910	100ML	31-JUL-2024		
2	K8J3M5	Ascorbic Acid	ST 5902	1G	01-AUG-2024		
3	B6N2P4	Magnesium Stearate	ST 5928	22.5L	09-SEP-2025		
4	Q1W5E7	Microcrystalline Cellulose	ST 3839	4L	10-NOV-2024		
5	X9C4V6	Ethanol	ST 2600	12G	01-JUN-2024		
6	G3T8H2	Sodium Chloride	ST 7737	1G	04-JUN-2024		
7	Z7Y6U1	Lactose Monohydrate	ST 1257	3ML	19-SEP-2025		
8	D5F1R9	Polyethylene Glycol	ST 8562	1G	25-SEP-2025		
9	L2K6J4	Propylene Glycol	ST 4358	100ML	12-DEC-2024		
10	M4N9B8	Hydroxypropyl Methylcellulose	ST 7516	1ML	17-FEB-2024	20-Aug-23	20-Aug-23
11	E7Q1W5	Croscarmellose Sodium	ST 9767	2L	11-JAN-2025		
12	V6X2C3	Povidone	ST 1162	1G	31-DEC-2024		
13	H9G4T7	Sodium Lauryl Sulfate	ST 6661	10ML	10-NOV-2024		
14	U1Z8Y5	Dimethyl Sulfoxide (DMSO)	ST 6518	1ML	12-OCT-2024		
15	P3D6F2	Gelatin	ST 2287	1G	21-JAN-2024		
16	R5A7E9	Polyvinyl Alcohol (PVA)	ST 7663	21.2L	23-AUG-2024		
17	T7H2J6	Stearic Acid	ST 5738	100ML	29-SEP-2024		
18	Y4V8K1	Titanium Dioxide	ST 2610	5L	04-JUN-2024		

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Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05- 9700
	Document Version Number: 2.3

1. Bill Of Material

Serial Number	Material ID	Material Description	Lot Number	Quantity	Expiry Date	Issued By	Received By
1	A2R7F9	Acetaminophen	ST 9124	90ML	03-NOV-2024		
2	K8J3M5	Ascorbic Acid	ST 6621	1.2G	01-DEC-2024		
3	B6N2P4	Magnesium Stearate	ST 9429	20.5L	26-JAN-2026		
4	Q1W5E7	Microcrystalline Cellulose	ST 4091	3L	11-MAR-2025		
5	X9C4V6	Ethanol	ST 3014	11G	04-OCT-2025		
6	G3T8H2	Sodium Chloride	ST 3192	90MG	21-OCT-2025		
7	Z7Y6U1	Lactose Monohydrate	ST 2.108	1ML	28-JAN-2024		
8	D5F1R9	Polyethylene Glycol	ST 9148	2G	17-JAN-2026		
9	L2K6J4	Propylene Glycol	ST 6365	100ML	16-MAR-2024		
10	M4N9B8	Hydroxypropyl Methylcellulose	ST 8437	2ML	11-JUN-2025	22-Aug-23	22-Aug-23
11	E7Q1W5	Croscarmellose Sodium	ST 3924	3L	04-MAY-2025		
12	V6X2C3	Povidone	ST 3140	1.5G	07-MAR-2025		
13	H9G4T7	Sodium Lauryl Sulfate	ST 6733	12ML	26-MAR-2025		
14	U1Z8Y5	Dimethyl Sulfoxide (DMSO)	ST 7692	1ML	18-FEB-2024		
15	P3D6F2	Gelatin	ST 3456	2G	23-MAY-2024		
16	R5A7E9	Polyvinyl Alcohol (PVA)	ST 5233	20L	24-DEC-2024		
17	T7H2J6	Stearic Acid	ST 4881	140ML	16-JAN-2025		
18	Y4V8K1	Titanium Dioxide	ST 3863	2L	24-OCT-2025		

3. In Sample-9690.pdf, the 'Equipment List' is present on Page No. 3, and in Sample-9700.pdf, the 'Equipment List' is present on Page No. 4. Map these 2 pages and check if the Equipment Number is the same or not for Equipment's - Gas Chromatography-Mass Spectrometry (GC-MS), Differential Scanning Calorimeter (DSC), Dissolution Tester. Flags: Yes/No

Note: You can save results in a tabular/dictionary/JSON

Ref: Snippet

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

2. Equipment List

Serial Number	Equipment	Manufacturer	Equipment Number	Experiemnt Used	Initials
1	High-Performance Liquid Chromatography (HPLC) System	Thermo Fisher Scientific	HPLC-05	✓Yes No NA Caliberation Due Date : 31-Dec-2025	
2	Gas Chromatography-Mass Spectrometry (GC-MS)	Thermo Fisher Scientific	GCMS-A11	✓Yes No NA Caliberation Due Date : 31-Dec-2025	
3	Fourier-Transform Infrared (FTIR) Spectrometer	Thermo Fisher Scientific		Yes ✓No NA Caliberation Due Date :	
4	Ultraviolet-Visible (UV-Vis) Spectrophotometer	Thermo Fisher Scientific	UVVIS-07	✓Yes No NA Caliberation Due Date : 7-June-2026	
5	Liquid Chromatography-Mass Spectrometry (LC-MS)	Waters Corporation	LSCM-09	✓Yes No NA Caliberation Due Date : 9-Jan-2028	
6	Differential Scanning Calorimeter (DSC)	Waters Corporation	DSC-05	✓Yes No NA Caliberation Due Date : 9-Jan-2024	
7	Nuclear Magnetic Resonance (NMR) Spectrometer	PerkinElmer		Yes ✓No NA Caliberation Due Date :	
8	Particle Size Analyzer	PerkinElmer		Yes ✓No NA Caliberation Due Date :	
9	Dissolution Tester	PerkinElmer	DT-005	✓Yes No NA Caliberation Due Date : 11-June-2026	

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05- 9700
	Document Version Number: 2.3

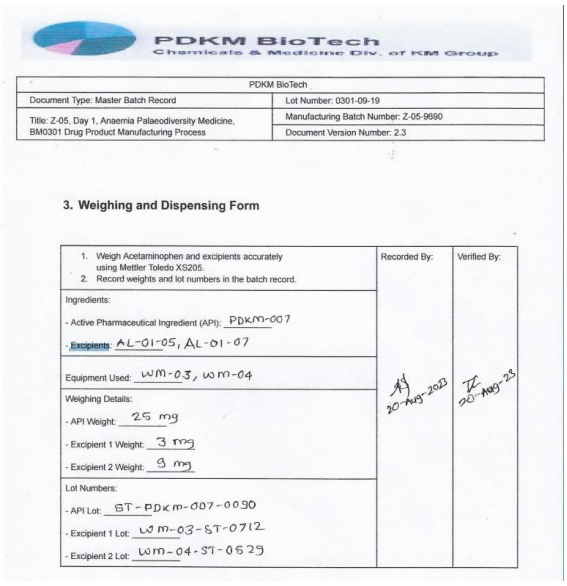
2. Equipment List

Serial Number	Equipment	Manufacturer	Equipment Number	Experiemnt Used	Initials
1	High-Performance Liquid Chromatography (HPLC) System	Thermo Fisher Scientific	HPLC-05	✓Yes No NA Caliberation Due Date : 31-Dec-2025	
2	Gas Chromatography-Mass Spectrometry (GC-MS)	Thermo Fisher Scientific	GCMS-A11	✓Yes No NA Caliberation Due Date : 31-Dec-2025	
3	Fourier-Transform Infrared (FTIR) Spectrometer	Thermo Fisher Scientific		Yes ✓No NA Caliberation Due Date :	
4	Ultraviolet-Visible (UV-Vis) Spectrophotometer	Thermo Fisher Scientific	UVVIS-07	✓Yes No NA Caliberation Due Date : 7-June-2026	
5	Liquid Chromatography-Mass Spectrometry (LC-MS)	Waters Corporation	LSCM-09	✓Yes No NA Caliberation Due Date : 9-Jan-2028	
6	Differential Scanning Calorimeter (DSC)	Waters Corporation	DSC-05	✓Yes No NA Caliberation Due Date : 9-Jan-2024	
7	Nuclear Magnetic Resonance (NMR) Spectrometer	PerkinElmer		Yes ✓No NA Caliberation Due Date :	
8	Particle Size Analyzer	PerkinElmer		Yes ✓No NA Caliberation Due Date :	
9	Dissolution Tester	PerkinElmer	DT-005	✓Yes No NA Caliberation Due Date : 11-June-2026	

4. In Sample-9690.pdf, the 'Weighing and Dispensing Form' is present on Page No. 4, and in Sample-9700.pdf on Page 8, the 'Equipment List' is present on Page No. 8. Map these 2 pages and check if the Ingredients (Active Pharmaceutical Ingredient (API), Excipients) are the same or not (Flags: Yes/No). If the same (Flags = Yes), then determine and save the CHANGE in Weighing Details (API Weight, Excipient 1 Weight, Excipient 2 Weight)

Note: You can save results in a tabular/dictionary/JSON

Ref: Snippet

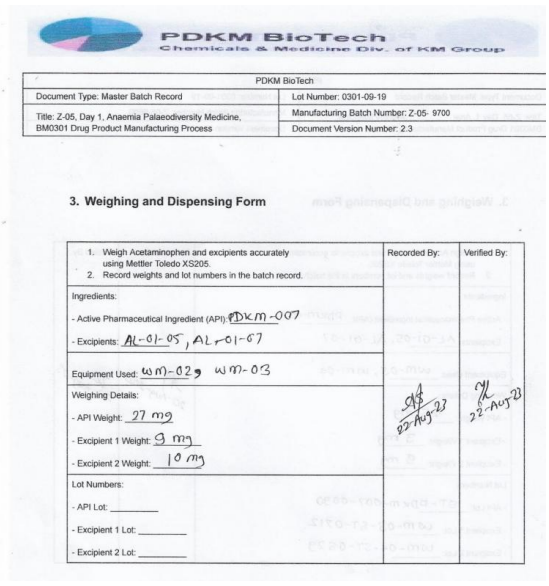


PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

3. Weighing and Dispensing Form

1. Weigh Acetaminophen and excipients accurately using Mettler Toledo XS205. 2. Record weights and lot numbers in the batch record.	Recorded By:	Verified By:
Ingredients: - Active Pharmaceutical Ingredient (API): PDKM-007 - Excipients: AL-01-05, AL-01-07		
Equipment Used: WM-03, WM-04		
Weighing Details: - API Weight: 25 mg - Excipient 1 Weight: 3 mg - Excipient 2 Weight: 9 mg	20-Aug-2023	20-Aug-23
Lot Numbers: - API Lot: ST-PDKM-007-0090 - Excipient 1 Lot: WM-03-ST-0712 - Excipient 2 Lot: WM-04-ST-0529		



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PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05- 9700
	Document Version Number: 2.3

3. Weighing and Dispensing Form

1. Weigh Acetaminophen and excipients accurately using Mettler Toledo XS205. 2. Record weights and lot numbers in the batch record.	Recorded By:	Verified By:
Ingredients: - Active Pharmaceutical Ingredient (API): PDKM-007 - Excipients: AL-01-05, AL-01-07		
Equipment Used: WM-02, WM-03		
Weighing Details: - API Weight: 27 mg - Excipient 1 Weight: 9 mg - Excipient 2 Weight: 10 mg	22-Aug-23	22-Aug-23
Lot Numbers: - API Lot: _____ - Excipient 1 Lot: _____ - Excipient 2 Lot: _____		

5. In Sample-9690.pdf, the 'Granulation' is present on Page No. 5, and in Sample-9700.pdf, the 'Granulation' is present on Page No. 5. Map these 2 pages and check if the Equipment Used is the same or not (Flags: Yes/No). If the same (Flags = Yes), then determine and save the CHANGE in Granulation Parameters (Mixing Time, Granulation Time)

Note: You can save results in a tabular/dictionary/JSON

Ref: Snippet

PDKM BioTech
Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05-9690
	Document Version Number: 2.3

4. Granulation

	Recorded By:	Verified By:
1. Combine weighed materials in Glatt GPCG Pro-5. 2. Add binder solution gradually while mixing. 3. Granulate the mixture to form uniform wet mass.		
Equipment Used: GL-003		
Binder Solution Details: - Binder Solution Used: BS-05 - Volume Added: 50mL	20-Aug-2023	20-Aug-23
Granulation Parameters: - Mixing Time: 59 sec - Granulation Time: 360 sec		

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Chemicals & Medicine Div. of KM Group

PDKM BioTech	
Document Type: Master Batch Record	Lot Number: 0301-09-19
Title: Z-05, Day 1, Anaemia Palaeodiversity Medicine, BM0301 Drug Product Manufacturing Process	Manufacturing Batch Number: Z-05- 9700
	Document Version Number: 2.3

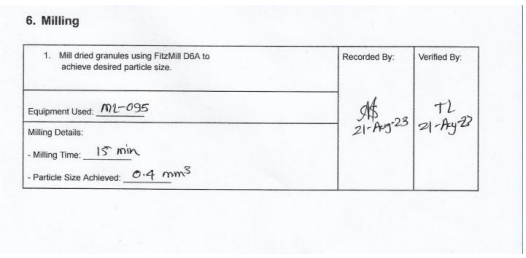
4. Granulation

	Recorded By:	Verified By:
1. Combine weighed materials in Glatt GPCG Pro-5. 2. Add binder solution gradually while mixing. 3. Granulate the mixture to form uniform wet mass.		
Equipment Used: GL-003		
Binder Solution Details: - Binder Solution Used: BS-05 - Volume Added: 50 mL	22-Aug-23	22-Aug-23
Granulation Parameters: - Mixing Time: 63 sec - Granulation Time: 400 sec		

6. In Sample-9690.pdf, the 'Milling' is present on Page No. 6, and in Sample-9700.pdf, the 'Milling' is present on Page No. 7. Map these 2 pages and save the handwritten text found in Sample-9700.pdf underneath the table.

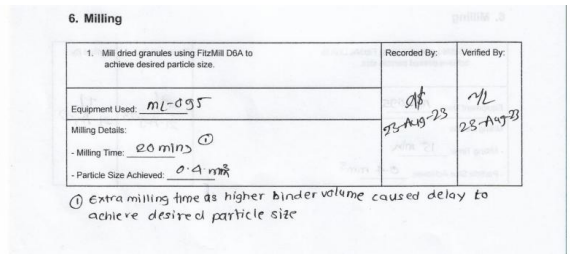
Note: You can save results in a tabular/dictionary/JSON

Ref: Snippet



6. Milling

1. Mill dried granules using FitzMill D6A to achieve desired particle size.	Recorded By:	Verified By:
Equipment Used: ML-095	AB 21-Aug-23	TL 21-Aug-23
Milling Details: - Milling Time: 15 min - Particle Size Achieved: 0.4 mm³		



6. Milling

1. Mill dried granules using FitzMill D6A to achieve desired particle size.	Recorded By:	Verified By:
Equipment Used: ML-095	AB 23-Aug-23	TL 23-Aug-23
Milling Details: - Milling Time: 20 mins ① - Particle Size Achieved: 0.4 mm³		

① Extra milling time as higher binder volume caused delay to achieve desired particle size